

Population pharmacokinetics of fluconazole in critically ill patients receiving extracorporeal membrane oxygenation and continuous renal replacement therapy: an ASAP ECMO study

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ABSTRACT This multicenter study describes the population pharmacokinetics (PK) of fluconazole in critically ill patients receiving concomitant extracorporeal membrane oxygenation (ECMO) and continuous renal replacement therapy (CRRT) and includes an evaluation of different fluconazole dosing regimens for achievement of target exposure associated with maximal efficacy. Serial blood samples were obtained from critically ill patients on ECMO and CRRT receiving fluconazole. Total fluconazole concentrations were measured in plasma using a validated chromatographic assay. A population PK model was developed and Monte Carlo dosing simulations were performed using Pmetrics in R. The probability of target attainment (PTA) of various dosing regimens to achieve fluconazole area under the curve to minimal inhibitory concentration ratio (AUC_{0-24}/MIC) >100 was estimated. Eight critically ill patients receiving concomitant ECMO and CRRT were included. A two-compartment model including total body weight as a covariate on clearance adequately described the data. The mean ($\pm$ standard deviation, SD) clearance and volume of distribution were 2.87 ± 0.63 L/h and 15.90 ± 13.29 L, respectively. Dosing simulations showed that current guidelines (initial loading dose of 12 mg/kg then 6 mg/kg q24h) achieved >90% of PTA for a MIC up to 1 mg/L. None of the tested dosing regimens achieved 90% of PTA for MIC above 2 mg/L. Current fluconazole dosing regimen guidelines achieved >90% PTA only for *Candida* species with MIC <1 mg/L and thus should be only used for *Candida*-documented infections in critically ill patients receiving concomitant ECMO and CRRT. Total body weight should be considered for fluconazole dose.

KEYWORDS extracorporeal membrane oxygenation, continuous renal replacement therapy, pharmacokinetics, fluconazole, critically ill patients

Critically ill patients often manifest extreme inter- and intraindividual pharmacokinetic (PK) variability (1). The use of extracorporeal support devices, namely continuous renal replacement therapy (CRRT) and extracorporeal membrane oxygenation (ECMO), may exacerbate this PK variability (2). Both devices are frequently used concomitantly in critically ill patients (3) and have different consequences on drug PK depending on drugs' physicochemical properties. Drugs that are hydrophilic, have low protein-binding properties, and possess small molecular weights are likely to be significantly dialyzed during CRRT (4). However, ECMO, because of the large volume of fluids (crystalloid, albumin, or exogenous blood) required to prime the ECMO circuit and the circuit's sequestration propensity, can also affect the volume of distribution and clearance of hydrophilic drugs (5, 6). Therefore, concomitant use of ECMO and CRRT is thought to lead to unpredictable drug concentrations exposing patients to under- or

Editor James E. Leggett, Providence Portland Medical Center, Portland, Oregon, USA

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The authors declare no conflict of interest.

See the funding table on p. 12.

Received 20 September 2023

Accepted 7 November 2023

Published 8 December 2023

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overdosage of hydrophilic drugs which in the case of antifungals could lead to treatment failure, antifungal resistance, or drug toxicity (2).

During the last decade, both venoarterial (7) and venovenous ECMO (8) have been increasingly used in critically ill patients to support refractory cardiovascular and/or respiratory failure. The clinical course of ECMO patients is commonly complicated by nosocomial infections, particularly fungal infections (9–11). During the recent coronavirus disease 2019 (COVID-19) pandemic, the use of ECMO was associated with a higher incidence of candidemia in critically ill patients (12). *Candida* infections, especially candidemia, were associated with increased mortality in patients receiving ECMO (13). Among the *Candida* spp., *Candida albicans* was the most encountered in ECMO patients with fungal infections (14).

Fluconazole is the first-line antifungal agent for the prophylaxis and treatment of *Candida albicans* infections (15, 16). Fluconazole exhibits concentration- and time-dependent antifungal activity with a prolonged post-antifungal effect, and the ratio of the area under the concentration-time curve during a 24-hour period to minimum inhibitory concentration (AUC_{0-24}/MIC) of >100 has been correlated with maximal clinical efficacy (17). Fluconazole is a hydrophilic (LogP of 0.5) drug with a low plasma protein binding property (12%) and possesses a low molecular weight (306.2 Da) (18). It is predominantly cleared by renal elimination, *via* both glomerular filtration and active tubular secretion (19). In critically ill adult patients receiving CRRT, the clearance of fluconazole is reported to increase relative to patients without renal dysfunction and therefore, higher dosing regimens are advocated to achieve exposures associated with maximal activity (20).

To date, except for one case report by Dhanani et al. (21), there is no study describing the PK of fluconazole in critically ill adult patients receiving ECMO. Therefore, there are no robust data to guide effective dosing in this specific population (5). The Antibiotic and Analgesic Pharmacokinetics during Extracorporeal Membrane Oxygenation (ASAP ECMO) study (22) was designed to describe the PK of commonly used drugs in critically ill adult patients receiving ECMO. The objective of the current analysis is to describe the population PK of fluconazole in critically ill adult patients receiving concomitant ECMO and CRRT and to identify dosing strategies that provide maximal exposures for clinical efficacy.

RESULTS

Population characteristics

In all, 10 patients were enrolled, two of whom did not receive CRRT and were excluded from the analysis. The final population included eight patients who contributed 67 plasma samples collected over one dosing interval. Table 1 summarizes the baseline demographic and clinical characteristics of the study population. The median (interquartile range; IQR) age was 43 years (41–49) and patients were predominantly male ($n = 6/8$; 75%). The median (IQR) total body weight was 95 kg (75–103). CVVHDF was the most common CRRT modality used in this study population ($n = 6/8$; 75%). Most patients ($n = 5/8$; 65%) received a dosing regimen of 800 mg as an initial loading dose on Day 1, followed by 400 mg q24h. The median (IQR) minimum (trough) drug concentration (C_{min}) was 6.8 (5.7–7.9) mg/L and the maximum drug concentration (C_{max}) was 16.6 (12.1–19.9) mg/L.

PK model building

The final population PK model that adequately described the plasma concentration-time data was a two-compartment model with total body weight as a covariate on CL (normalized to the median total body weight of 95 kg). Neither ECMO-related nor CRRT-related variables significantly improved the PK model. The following equation is the mathematical calculation used to generate the individual TBW-scaled CL:

$$CL_{tbw} = CLx\left(\frac{tbw}{95}\right)^{0.75}$$

Table 2 presents the PK parameter estimates of the final population PK model. Figure 1 shows the diagnostic plots for the final model for total fluconazole serum concentrations. The model produced acceptable fits in the diagnostic plots (Fig. 2).

TABLE 1 Baseline characteristics, clinical information of the studied population, and relevant ECMO and RRT data^a

Demographic and ECMO data	Result ^a (n = 8)
Male	6 (75)
Age (years)	43 [41-49]
Total body weight (kg) at admission	95 [75-102]
Ideal body weight (kg)	73 [69-81]
Adjusted body weight (kg)	90 [72-94]
Lean body weight (kg)	65 [59-73]
Body mass index (kg/m ²)	27.2 [23.4–31.6]
Body surface area (m ²)	2.1 [1.9–2.3]
Indication for ECMO:	
Cardiogenic shock	2 (25)
Heart transplantation	2 (25)
Lung transplantation	2 (25)
Cardiomyopathy	1 (12.5)
Pneumonia	1 (12.5)
Type of ECMO:	
VA ECMO	4 (50)
VV ECMO	3 (38)
VA and VV ECMO	1 (12)
ECMO Flow (L/min)	3.5 [2.7–3.7]
Days under ECMO before sampling	9 [6-12]
ECMO pump:	
Jostra RotaFlow	5 (62)
Cardiohelp	3 (38)
ECMO oxygenator:	
Maquet Quadrox	8 (100)
Illness severity score:	
APACHE II score (on admission)	18 [13-22]
SOFA score (on sampling day)	8 [7-16]
CRRT mode:	8 (100)
CVVH	1 (12.5)
CVVHD	1 (12.5)
CVVHDF	5 (75)
Effluent rate (mL/kg/h) (n = 7)	37.6 [35.5–39.6]
Blood flow rate (mL/min) (n = 7)	200 [200-250]
Urea (mmol/L)	11.7 [10.5–16.0]
Albumin (g/L)	29 [20-33]
Bilirubin (mmol/L)	15 [13-21]
No. of patients by dosing regimen administered:	
800 mg (8–10 mg/kg) ^b Day 1 then 400 mg (4–5 mg/kg) ^b q24h	5 (62)
400 mg (6 mg/kg) ^b Day 1 then 400 mg (6 mg/kg) ^b q24h	2 (25)
100 mg (1 mg/kg) ^b Day 1 then 100 mg (6 mg/kg) ^b q24h	1 (13)

^aData displayed in median [IQR, interquartile range] or n (%), as appropriate.

^bBased on total body weight. The non-fixed dose (mg) reports the intended dose as it was prescribed, and the weight-based dose (mg/kg) reports the actual dose.

^cVA, venous arterial ECMO; VV, venovenous ECMO; APACHE II, Acute Physiology and Chronic Health Evaluation II; CRRT, continuous renal replacement therapy; CVVH, continuous venovenous hemofiltration; CVVHD, continuous venovenous hemodialysis; CVVHDF continuous venovenous hemodiafiltration; SOFA, Sequential Organ Failure Assessment.

TABLE 2 Pharmacokinetic parameter estimates for the final fluconazole pharmacokinetic model^a

Parameter	Mean	SD	CV%	Median	95% CI	Shrinkage (%)
CL (L/h)	2.87	0.63	21.84	2.6	2.38–3.46	0.12
V _c (L)	15.90	13.29	83.55	10.38	3.74–34.05	0.00
Q (L/h)	84.73	48.60	57.36	103.77	32.29–147.77	0.25
V _p (L)	36.06	13.93	38.63	38.59	22.38–49.80	0.01

^a95% CI, 95% bootstrapped confidence interval; CL, fluconazole clearance; V_c, volume of distribution of the central compartment; V_p, volume of distribution of the peripheral compartment; Q, intercompartmental clearance.

Dosing simulations

Figure 3 and 4 show the PTAs for AUC_{0–24}/MIC > 100 at Days 1 and 7 of various fluconazole dosing regimens across a range of MICs and various total body weights. All dosing regimens achieved 100% of PTA against *Candida* spp. with an MIC ≤ 1 mg/L. For an MIC of up to 2 mg/L, only dosing regimens including a loading dose of 18 mg/kg achieved >90% of PTA regardless of the TBW. On Day 7, only increasing the maintenance dose to 12 mg/kg q24h or reducing the dosing interval to q12h achieved >90% of PTA for an MIC up to 2 mg/L. None of the tested dosing regimens achieved efficacious PTA for MIC > 2 mg/L.

The FTAs of all simulated fluconazole dosing regimens over three different TBW and at Days 1 and 7 were ≥96% against the empirical MIC distribution of *C. albicans* and 100% against the directed MIC distribution (Table 3).

DISCUSSION

To the best of our knowledge, this is the first paper to describe the population PK of fluconazole in critically ill adult patients receiving concomitant ECMO and CRRT. This study found that the currently recommended fluconazole dosing regimen of 12 mg/kg as an initial loading dose followed by 6 mg/kg q24h provided a high probability of efficacy only against *Candida* spp. with an MIC ≤ 1 mg/L. In *Candida* spp. which exhibits MICs of up to 2 mg/L, the use of a higher loading dose of 18 mg/kg and/or higher maintenance dose (12 mg/kg) and/or more frequent daily dosing (6 mg/kg q12h) should be recommended. We also found that total body weight (TBW) should be considered and used for weight-based dosing of fluconazole. It must be highlighted that, when using TBW for the loading dose calculation, the obese patients with a TBW greater than 100 kg, a 1,800 to 2,000 mg dose of fluconazole may be required. Very few data existed regarding fluconazole toxicity (23). Based on a small cohort of 28 patients, minor toxicity

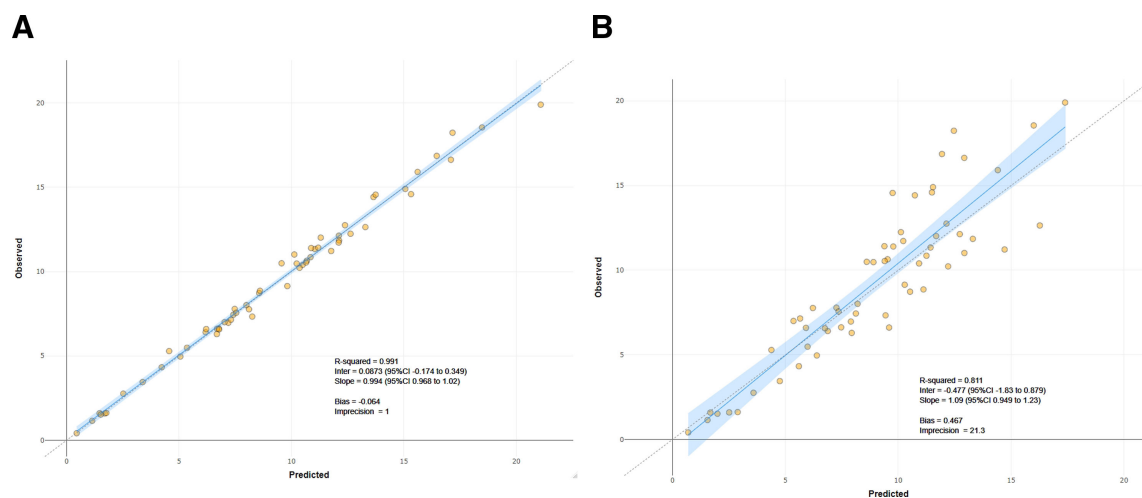


FIG 1 Diagnostic plots for the final model for total fluconazole serum concentrations. Data are presented in mg/L. (A) Observed versus individual predicted plots. (B) Population versus individual predicted plots

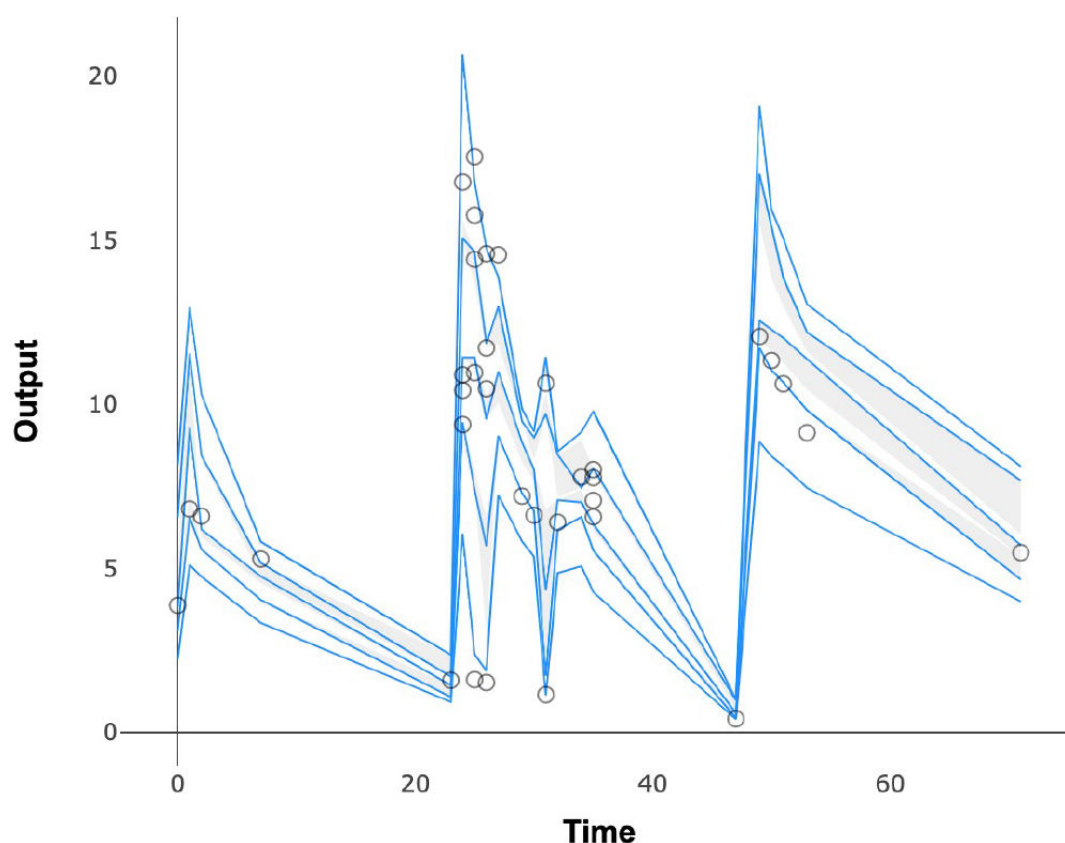


FIG 2 Visual predictive check of the final population PK model for total serum fluconazole concentrations.

(e.g., elevated liver function test results or confusion occurrence) has been reported in cases of doses greater than 2,000 mg (24). However, our simulation suggested only one initial high loading dose is required, at antifungal introduction. Thus, the likelihood of toxicity would be relatively low.

In the current study population where all patients were receiving both ECMO and CRRT, the PK of fluconazole was best described using a two-compartment model with TBW included as a covariate on CL. The central volume of distribution of fluconazole in this study was 15.90 L and the CL was 2.87 L/h. The CL was consistent with previously published data (20, 25), while the central volume of distribution was lower than the estimate reported in a previous study (15.90 vs 46.3 L (26)). It must be emphasized that the data regarding PK of fluconazole in patients receiving ECMO are scarce and come mostly from pediatric populations (27, 28). One study has been performed in 30 critically ill adult patients of which five were receiving ECMO and six CRRT (26). However, these PK studies used a one-compartment model which precludes compara-

TABLE 3 Empirical and directed FTA for evaluated dosing regimens against *C. albicans*^{a,b}

	Empirical						Directed
	TBW 75 kg		TBW 95 kg		TBW 102 kg		All TBW
	Day 1	Day 7	Day 1	Day 7	Day 1	Day 7	Days 1 and 7
LD 12 mg/kg then 6 mg/kg q24h	96.54	96.35	96.69	96.44	96.73	96.45	100.0
LD 18 mg/kg then 6 mg/kg q24h	96.94	96.35	97.20	96.44	97.25	96.45	100.0
LD 18 mg/kg then 12 mg/kg q24h	96.94	97.09	97.20	97.21	97.25	97.24	100.0
LD 12 mg/kg then 6 mg/kg q12h	96.54	97.15	96.72	97.29	96.79	97.30	100.0

^aLD: loading dose; TBW: total body weight.

^bThe fractional target attainment using intermittent administration for various fluconazole doses based on the MIC (EUCAST, accessed 30 October 2023) for *Candida albicans*. The FTA assessed whether simulated patients maintained fluconazole serum concentrations above the MIC at Day 1 and Day 7.

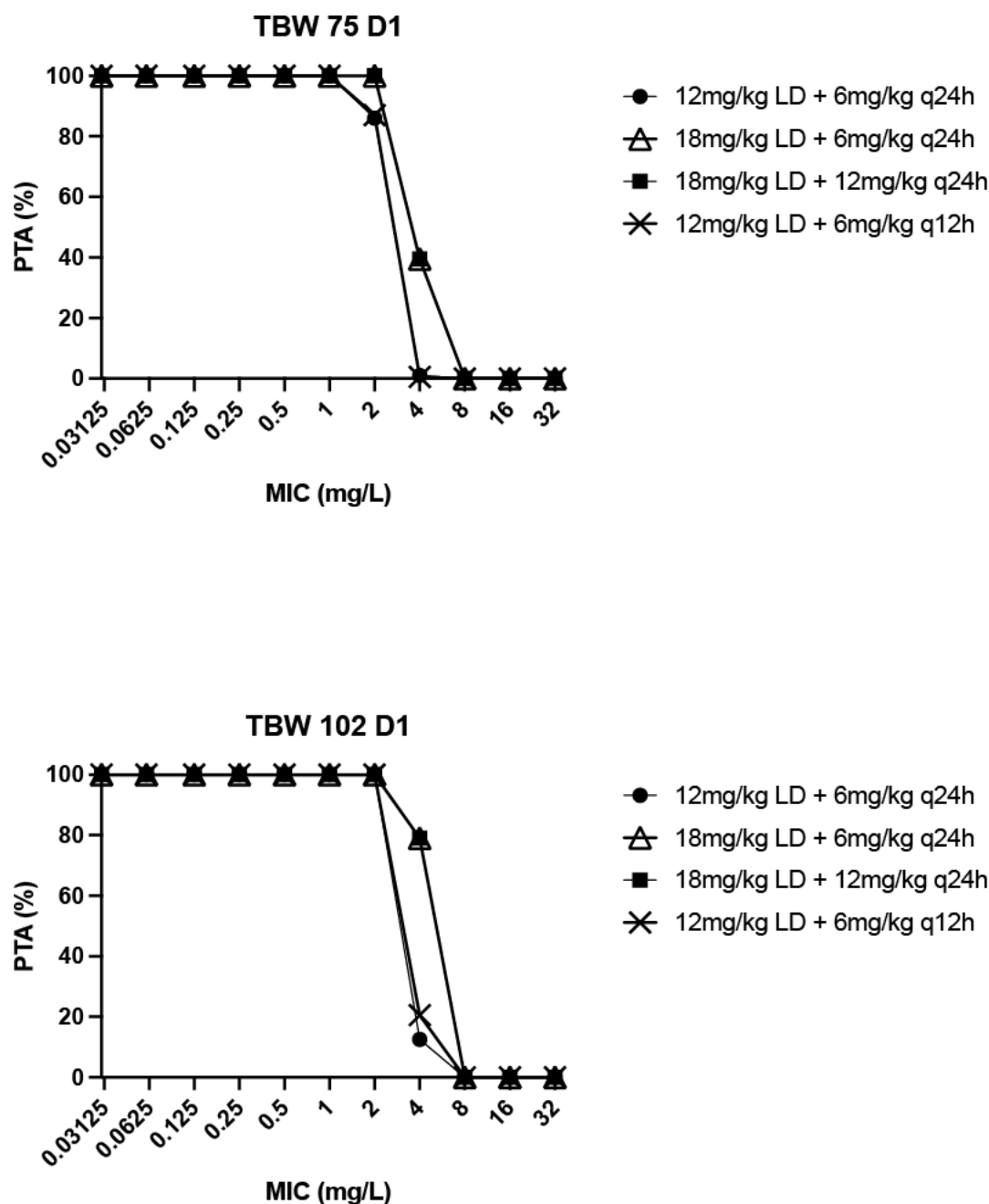


FIG 3 Monte Carlo simulations ($n = 1,000$) and probability of target attainment for achieving total serum fluconazole concentrations at target $AUC_{0-24}/MIC > 100$ at Day 1 for various dosing regimens according to the total body weight (TBW >75 kg and TBW >102 kg).

bility to our results. Lastly, one case report involving a critically ill patient receiving fluconazole while on ECMO and CRRT was published in 2021 (21). In this paper, the authors reported a higher volume of distribution (0.98 L/kg, ~40% higher) compared to patients not receiving ECMO. The PK differences between the current and previous studies can be potentially explained by the population that was included. All patients in this study were receiving both ECMO and CRRT. Seventy-five percent of the study cohort received the same CRRT modality (i.e., CVVHDF) and the CL was in accordance with previously published data (20). The difference in the central volume of distribution could be explained by two mechanisms. First, the specificity of the included cohort. Indeed, the most frequent indication for ECMO was cardiovascular insufficiency in different

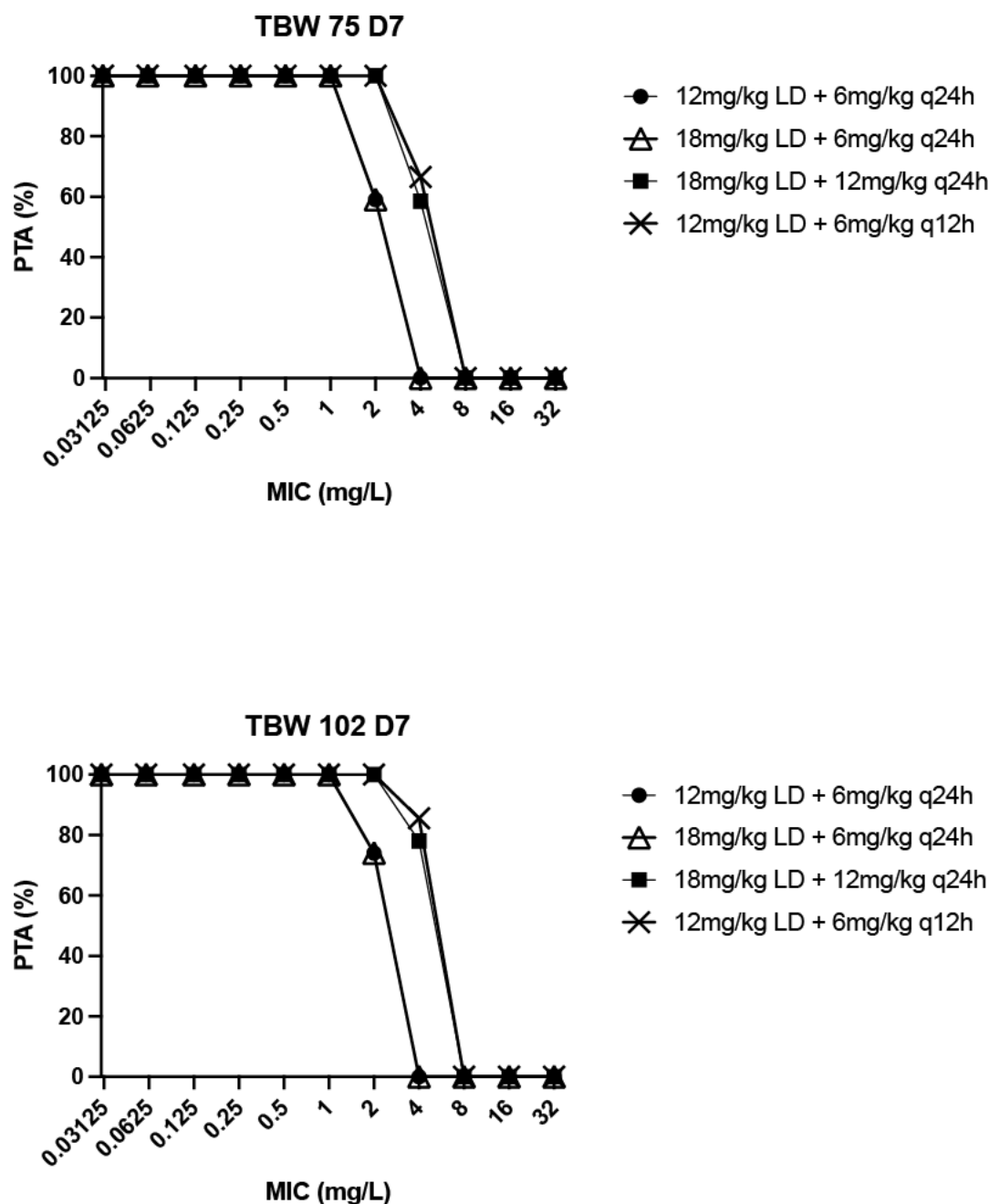


FIG 4 Monte Carlo simulations ($n = 1,000$) and probability of target attainment for achieving total serum fluconazole concentrations at target $AUC_{0-24}/MIC > 100$ at Day 7 for various dosing regimens according to the total body weight (TBW >75 kg and TBW >102 kg).

contexts with no septic shock, which could explain the lower volume of distribution in our study. Indeed, cardiovascular insufficiency is associated with less capillary leakage or restrictive vascular filling compared to septic shock (29). Second, the change in central volume of distribution because of ECMO is more likely to occur on the introduction of ECMO because of hemodilution from priming solution and ECMO-related physiological changes such as systemic inflammatory response syndrome (30). However, in our study, the median number of days under ECMO before sampling was 9 days (IQR 6–12), which was later than the early resuscitation phase.

During the pharmacokinetic model-building process, lean body weight and total body weight on CL improved the model. We decided to keep TBW as a covariate on CL

because it provided more accurate goodness-of-fit plots, bias, and imprecision parameters, and the difference between models including LBW and TBW were not significantly different (please refer to supplementary materials). Moreover, TBW was already identified as a significant covariate. Indeed, a previous population PK study of fluconazole in critically ill obese patients not receiving RRT demonstrated that the loading dose should be calculated based on total body weight (i.e., 12 mg/kg) while the maintenance dose should be based on renal clearance (31). Lastly, when performing PTA with both models, the results were relatively better with TBW than LBW incorporation. As current guidelines recommend weight-based dosing for fluconazole (15, 32) and TBW is easier used during routine clinical practice, we kept TBW instead of LBW.

One of the study's originalities was that we have combined two extracorporeal modalities (CRRT and ECMO) that could both influence the PK of fluconazole, at different degrees, at least in theory. Of note, it is likely that the PK of fluconazole would be significantly influenced by the CRRT than the ECMO by itself (18). When constructing the PK model, ECMO-related variables (e.g., ECMO types, flow rates, days under ECMO) were not identified as significant predictors of the primary PK parameters. Conversely, when looking at CRRT-related variables, the effluent rate and RRT mode were correlated with CL, with an R^2 of 0.33 and 0.42, respectively. However, when we added these covariates to the structural model, the model did not significantly improve statistically. This could be explained potentially by the small sample size and limited RRT data. The use of CRRT has been shown to influence fluconazole PK in the literature (4, 25, 26, 33, 34). According to Li et al., CRRT modalities and effluent flow rates should be considered during fluconazole dosing adjustment and a higher dose and/or shorter dosing interval must be used, especially when hemodialysis is used (4). The recent study by Roberts et al. highlighted that combining convection and diffusion (i.e., CVVHDF) resulted in greater drug CL than by convection or diffusion alone (35). Therefore, they recommended a loading dose of 12 mg/kg followed by 6 mg/kg q12h as a maintenance dose in patients receiving CVVHDF. In our study population, the CVVHDF modality was most frequently used. Our dosing simulations showed that 6 mg/kg q12h was one of two dosing regimens able to demonstrate 100% target attainment at Day 7 up to a MIC of 2 mg/L. Therefore, our results are currently inline with recent data concerning the fluconazole dosing regimen during RRT (26, 36).

The current fluconazole dosing guideline for critically ill patients recommends a loading dose of 12 mg/kg followed by 6 mg/kg q24h for patients who have had no recent azole exposure and are not colonized with azole-resistant *Candida* spp. (15, 31). In the case of CRRT, recent studies suggest keeping the same loading dose but in increasing the maintenance dose to 6 mg/kg q12h (35). In the present study, all dosing regimens demonstrated more than 90% of PTA only for an MIC of up to 1 mg/L, and 100% of directed FTA against *C. albicans*. Thus, in the case of local ecology with a high prevalence of *Candida albicans* infection, the current dosing regimen is efficacious and safe and avoids unnecessary echinocandin exposure. However, in the case of local ecology with *Candida* spp. exhibiting MIC of up to 2 mg/L, only increasing the maintenance dose or reducing the interval between two dosing achieved the pre-defined target. Therefore, in such a situation with a high prevalence of *Candida* spp. exhibiting an MIC of >2 mg/L (i.e., *C. glabrata*), we suggest using a different class of antifungal agent.

This study has several limitations. First, this study did not identify ECMO- or CRRT-related variables to be significant predictors of fluconazole clearance. This could be explained by the small sample size and the distribution of PK concentrations (mostly during the first 36 hours), and also by the limited data regarding CRRT. A larger sample size and more PK data are required to better describe the PK variability of this population. Nevertheless, our cohort could be considered as homogeneous population of critically ill adult patients on ECMO and CRRT and could be assumed as a good starting point for dosing regimen recommendations. Second, samples were obtained from an arterial line and not from the ECMO or the CRRT circuit. When sampling directly from the circuit in parallel to the arterial line, important information could be obtained to

better describe fluconazole PK. Lastly, no information on the type of infection, candida isolation, empirical versus prophylaxis versus directed indication for fluconazole use, and patient outcomes was collected in this study. Therefore, the proposed fluconazole dosing regimens should now be confirmed in clinical studies with outcomes such as clinical and/or microbiological cure, ICU/hospital mortality, and length of stay being evaluated.

Conclusion

This is the first study to describe the PK of fluconazole in critically ill adult patients on ECMO and CRRT. The current dosing regimen of fluconazole (a loading dose of 12 mg/kg followed by 6 mg/kg q24h) in patients on ECMO and RRT provides a high PTA against *Candida* spp. with an MIC of up to 1 mg/L. Based on our model and simulations, TBW should be considered and used to calculate fluconazole doses.

MATERIALS AND METHODS

Setting

The ASAP ECMO study was a prospective, open-labeled, multi-center PK study conducted in five intensive care units (ICU) across Australia, New Zealand, South Korea, and Switzerland from November 2012 to November 2019. A detailed study protocol has been published elsewhere and is briefly discussed here (22). Ethical approval was provided by the lead site (The Prince Charles Hospital, Brisbane, Australia, HREC/11/QPCH/121) with individual institutional approvals obtained according to local protocols: Auckland City Hospital, Auckland, New Zealand—LRS/12/06/020/AM08; The Alfred Hospital, Melbourne, Australia—541/12, Bern University Hospital, Bern, Switzerland—2017-01315; Seoul National University Bundang Hospital, Seoul, Republic of Korea—B-1804/465-305; and St Vincent's Hospital, Sydney, Australia—HREC/11/SVH/97. Written informed consent was acquired from each participant's next of kin.

Study population

Patients between 18 and 90 years, hospitalized in ICU, and receiving fluconazole whilst undergoing ECMO for respiratory and/or cardiac dysfunction were eligible for inclusion. Patients who had a known allergy to the study drug, who were pregnant, who had bilirubin >150 $\mu\text{mol/L}$, received ongoing massive blood transfusion (>50% blood volume) in the previous 8 hours or who received therapeutic plasma exchange in the previous 24 hours were excluded.

Fluconazole dosing and administration

Fluconazole was reconstituted and administered intravenously as an intermittent infusion, according to local hospital protocols.

Study procedures/protocol

PK sampling was performed on one or two dosing occasions after the patient was stabilized on ECMO (i.e., not receiving ongoing acute resuscitation). Blood samples were drawn from an existing arterial line and collected into 2 mL lithium-heparinized tubes at 0, 15, 30, 45, 60, 90, 120, 180, 360, and 480 minutes post-commencement of infusion. Blood samples were centrifuged at 3,000 g for 10 minutes to separate plasma. Plasma samples were frozen at the study site at -80°C . All frozen plasma samples were then couriered to and assayed at the central bioanalysis laboratory at the University of Queensland Centre for Clinical Research (UQCCR), Brisbane, Australia. De-identified clinical and demographic data were collected by trained research staff. Each study site maintained an electronic database for its participants, which was subsequently consolidated into a single database.

Fluconazole assay

Total fluconazole concentrations were measured in plasma by a validated ultra-high-performance liquid chromatography using two mass spectrometry detectors (UHPLC-MS/MS) on a Nexera2-8030+ system (Shimadzu, Kyoto, Japan). Clinical samples were assayed in batches alongside calibrators and quality controls, and the results were subjected to batch acceptance criteria. Linearity was validated over the concentration range of 0.1–50 mg/L. Validation of fluconazole measurement demonstrated precision within 4.3% and accuracy within 3.6% at 0.1, 0.3, 2, and 16 mg/L.

Population pharmacokinetic analysis

Pmetrics version 2.2.1 (Laboratory of Applied Pharmacokinetics and Bioinformatics, Los Angeles, CA, USA) for R (version 4.3.1) was used to develop a population PK model for fluconazole (37). One-, and two-compartment models were first constructed using non-parametric adaptative grid (NPAG) algorithms with total plasma fluconazole concentrations. A stepwise approach was undertaken to build the fluconazole PK model to establish: (i) the structural base model, (ii) the best-fit error model, and (iii) the development of a covariate model. Elimination from the central compartment was modeled as a first-order process. Lambda (additive) and gamma (multiplicative) error models were evaluated.

The effect of several biologically plausible demographic and clinical characteristics on the PK of fluconazole was assessed for linear, exponential, or logarithmic relationships and tested for inclusion into the model. The covariates tested for inclusion were age, gender, total body weight, body surface area, body mass index, ideal body weight, adjusted body weight, lean body weight, urea, bilirubin, albumin, ECMO mode, ECMO flow rate, days on ECMO, CRRT modality (continuous venovenous hemodialysis, CVVHD; continuous venovenous hemofiltration, CVVH; and continuous venovenous hemodiafiltration, CVVHDF), CRRT effluent flow rate, CRRT blood flow rate, Acute Physiology and Chronic Health Evaluation II (APACHE II), and Sequential Organ Failure Assessment (SOFA) scores. As all patients received RRT, renal clearance was not calculated.

Population pharmacokinetic model diagnostic

The best structural model was chosen according to the goodness-of-fit of both the population- and individual-observed versus predicted concentration plots, the value of the comparative log-likelihood ($-2*LL$), and the Akaike information criterion (AIC) of each run. The goodness-of-fit of the model was assessed by visual inspection of the observed versus predicted concentration plots both in population- or individual-scatter plots, the coefficient of determination (r^2), slopes, intercept, and bias of the linear regression. The predictive performance was also assessed for mean prediction error (bias) and the mean bias-adjusted squared (imprecision) of the population and individual posterior predictions. Internal validation was conducted *via* a visual predictive check (VPC) to evaluate the predictive performance of the final model with covariates using 1,000 simulations. The distribution of the observed concentration in this simulation was plotted and visually inspected. The final estimated PK parameters from the final model were presented as mean, standard deviation (SD), percentage coefficient of variation (%CV), median value, and percentage of shrinkage. The %CV represented the inter-individual variability. A bootstrap was performed by Pmetrics using a weighted marginal distribution of each population parameter, with the 95% confidence interval of the median calculated as a measure of the variance as a normal distribution.

Monte Carlo dosing simulations

Different fluconazole dosing regimens in patients with various total body weights (< 75 kg, 75 kg, >102 kg) were evaluated using Monte Carlo dosing simulation in Pmetrics ($n = 1,000$) using the final model against a specific *Candida* spp. MIC value ranging from

0.03 to 32 mg/L. The ratio of $AUC_{0-24}/MIC >100$ was chosen as the efficacy target for species (38), taking into account the protein binding of fluconazole of 12%. The following dosing regimens were evaluated: (i) a loading dose of 12 mg/kg followed by 6 mg/kg q24h; (ii) a loading dose of 18 mg/kg followed by 6 mg/kg q24h; (iii) a loading dose of 18 mg/kg followed by 12 mg/kg mg q24h; and (iv) a loading dose of 12 mg/kg followed by 6 mg/kg q12h. All PTA simulations were assessed at Day 1 (to evaluate the loading dose) and at steady state (defined as the seventh day of treatment). A PTA value of $\geq 90\%$ for a particular MIC was considered optimal in our study. To calculate the fractional target attainment (FTA), the PTA of each dosing regimen was compared against the MIC distribution of *Candida albicans* obtained from the European Committee on Antimicrobial Susceptibility Testing (EUCAST) database (available from www.eucast.org; Antimicrobial wild-type distributions of microorganisms, accessed 30 October 2023). The empirical FTA was calculated considering the whole range of MIC distribution of *Candida albicans* ($n = 2,175$) while the directed FTA was calculated by considering a range of MICs of the susceptible strains of *Candida albicans*. Any dosing regimen that achieved an FTA value of $\geq 95\%$ was considered an “optimal” dosing regimen either for empirical or directed fluconazole therapy.

As there is no published threshold and specific exposure parameters (e.g., AUC and C_{min}) that have been associated with fluconazole toxicity, we were not able to incorporate this evaluation into our dosing simulations (23). Anaissie et al. reported that more adverse effects were observed in patients receiving $\geq 1,600$ mg/day and these events were classified as “possibility related” to fluconazole (24). Therefore, as the dosing regimens evaluated in our study were all lower than 1600 mg/day, we assumed that the likelihood of toxicity would be relatively low.

Statistical analysis

Descriptive analyses were performed using Excel V16.71 where categorical variables are presented as counts (percentage) and continuous variables are presented as mean ($\pm$ SD).

ACKNOWLEDGMENTS

This work was supported by The Prince Charles Hospital Research Foundation, Australian and New Zealand Intensive Care Foundation, Australia and New Zealand College of Anaesthetists (S12/001; S13/021), Society of Hospital Pharmacists of Australia, National Health and Medical Research Council of Australia (APP1079421; APP1099452).

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FUNDING

Funder	Grant(s)	Author(s)
The Prince Charles Hospital Research Foundation; ANZIC Foundation; ANZ College of Anaesthetists		Kiran Shekar
SHPA; NHMRC Australia		Jason A. Roberts

ADDITIONAL FILES

The following material is available [online](#).

Supplemental Material

Supplemental material (AAC01201-23-S0001.docx). Supplemental materials 1, 2, 3, 4.

REFERENCES

- Roberts JA, Paul SK, Akova M, Bassetti M, De Waele JJ, Dimopoulos G, Kaukonen K-M, Koulenti D, Martin C, Montravers P, Rello J, Rhodes A, Starr T, Wallis SC, Lipman J, DALI Study. 2014. DALI: defining antibiotic levels in intensive care unit patients: are current β -lactam antibiotic doses sufficient for critically ill patients? *Clin Infect Dis* 58:1072–1083. <https://doi.org/10.1093/cid/ciu027>
- Roberts JA, Abdul-Aziz MH, Lipman J, Mouton JW, Vinks AA, Felton TW, Hope WW, Farkas A, Neely MN, Schentag JJ, Drusano G, Frey OR, Theuretzbacher U, Kuti JL, International Society of Anti-Infective Pharmacology and the Pharmacokinetics and Pharmacodynamics Study Group of the European Society of Clinical Microbiology and Infectious Diseases. 2014. Individualised antibiotic dosing for patients who are critically ill: challenges and potential solutions. *Lancet Infect Dis* 14:498–509. [https://doi.org/10.1016/S1473-3099\(14\)70036-2](https://doi.org/10.1016/S1473-3099(14)70036-2)
- Gu M, Mei X-L, Zhao Y-N. 2021. A review on extracorporeal membrane oxygenation and kidney injury. *J Biochem Mol Toxicol* 35:e22679. <https://doi.org/10.1002/jbt.22679>
- Li L, Li X, Xia Y, Chu Y, Zhong H, Li J, Liang P, Bu Y, Zhao R, Liao Y, Yang P, Lu X, Jiang S. 2020. Recommendation of antimicrobial dosing optimization during continuous renal replacement therapy. *Front Pharmacol* 11:786. <https://doi.org/10.3389/fphar.2020.00786>
- Lyster H, Shekar K, Watt K, Reed A, Roberts JA, Abdul-Aziz M-H. 2023. Antifungal dosing in critically ill patients on extracorporeal membrane oxygenation. *Clin Pharmacokinet* 62:931–942. <https://doi.org/10.1007/s40262-023-01264-0>
- Lyster H, Pitt T, Maunz O, Diamond S, Roberts JA, Brown D, Mills J, Armstrong-James D, Gerovasili V, Carby M, Dunning J, Simon A, Reed A. 2023. Variable sequestration of antifungals in an extracorporeal membrane oxygenation circuit. *ASAIO J* 69:309–314. <https://doi.org/10.1097/MAT.0000000000001802>
- Karagiannidis C, Brodie D, Strassmann S, Stoelben E, Philipp A, Bein T, Müller T, Windisch W. 2016. Extracorporeal membrane oxygenation: evolving epidemiology and mortality. *Intensive Care Med* 42:889–896. <https://doi.org/10.1007/s00134-016-4273-z>
- Munshi L, Walkey A, Goligher E, Pham T, Uleryk EM, Fan E. 2019. Venovenous extracorporeal membrane oxygenation for acute respiratory distress syndrome: a systematic review and meta-analysis. *Lancet Respir Med* 7:163–172. [https://doi.org/10.1016/S2213-2600\(18\)30452-1](https://doi.org/10.1016/S2213-2600(18)30452-1)
- Li X, Wang L, Wang H, Hou X. 2022. Outcome and clinical characteristics of nosocomial infection in adult patients undergoing extracorporeal membrane oxygenation: a systematic review and meta-analysis. *Front Public Health* 10:857873. <https://doi.org/10.3389/fpubh.2022.857873>
- Austin DE, Kerr SJ, Al-Soufi S, Connellan M, Spratt P, Goeman E, Nair P. 2017. Nosocomial infections acquired by patients treated with extracorporeal membrane oxygenation. *Crit Care Resusc* 19:68–75.
- Cavayas YA, Yusuff H, Porter R. 2018. Fungal infections in adult patients on extracorporeal life support. *Crit Care* 22:98. <https://doi.org/10.1186/s13054-018-2023-z>
- Alessandri F, Ceccarelli G, Migliara G, Baccolini V, Russo A, Marzuillo C, Ceparano M, Giordano G, Tozzi P, Galardo G, Raponi G, Mastroianni C,

- Venditti M, Pugliese F, d'Ettorre G. 2023. High incidence of candidemia in critically ill COVID-19 patients supported by veno-venous extracorporeal membrane oxygenation: a retrospective study. *J Fungi (Basel)* 9:119. <https://doi.org/10.3390/jof9010119>
13. Poth JM, Schewe J-C, Putensen C, Ehrentraut SF. 2022. Impact of invasive fungal diseases on survival under veno-venous extracorporeal membrane oxygenation for ARDS. *J Clin Med* 11:1940. <https://doi.org/10.3390/jcm11071940>
 14. Sun H-Y, Ko W-J, Tsai P-R, Sun C-C, Chang Y-Y, Lee C-W, Chen Y-C. 2010. Infections occurring during extracorporeal membrane oxygenation use in adult patients. *J Thorac Cardiovasc Surg* 140:1125–32. <https://doi.org/10.1016/j.jtcvs.2010.07.017>
 15. Pappas PG, Kauffman CA, Andes DR, Clancy CJ, Marr KA, Ostrosky-Zeichner L, Reboli AC, Schuster MG, Vazquez JA, Walsh TJ, Zaoutis TE, Sobel JD. 2016. Clinical practice guideline for the management of Candidiasis: 2016 update by the infectious diseases society of America. *Clin Infect Dis* 62:e1–50. <https://doi.org/10.1093/cid/civ933>
 16. Martin-Loeches I, Antonelli M, Cuenca-Estrella M, Dimopoulos G, Einav S, De Waele JJ, Garnacho-Montero J, Kanj SS, Machado FR, Montravers P, Sakr Y, Sanguinetti M, Timsit J-F, Bassetti M. 2019. ESICM/ESCMID task force on practical management of invasive candidiasis in critically ill patients. *Intensive Care Med* 45:789–805. <https://doi.org/10.1007/s00134-019-05599-w>
 17. Clancy CJ, Yu VL, Morris AJ, Snyderman DR, Nguyen MH. 2005. Fluconazole MIC and the fluconazole dose/MIC ratio correlate with therapeutic response among patients with candidemia. *Antimicrob Agents Chemother* 49:3171–3177. <https://doi.org/10.1128/AAC.49.8.3171-3177.2005>
 18. Sinnollareddy M, Peake SL, Roberts MS, Lipman J, Roberts JA. 2012. Using pharmacokinetics and pharmacodynamics to optimise dosing of antifungal agents in critically ill patients: a systematic review. *Int J Antimicrob Agents* 39:1–10. <https://doi.org/10.1016/j.ijantimicag.2011.07.013>
 19. Bellmann R, Smuszkiwicz P. 2017. Pharmacokinetics of antifungal drugs: practical implications for Optimized treatment of patients. *Infection* 45:737–779. <https://doi.org/10.1007/s15010-017-1042-z>
 20. Patel K, Roberts JA, Lipman J, Tett SE, Deldot ME, Kirkpatrick CM. 2011. Population pharmacokinetics of fluconazole in critically ill patients receiving continuous venovenous hemodiafiltration: using monte carlo simulations to predict doses for specified pharmacodynamic targets. *Antimicrob Agents Chemother* 55:5868–5873. <https://doi.org/10.1128/AAC.00424-11>
 21. Dhanani JA, Lipman J, Pincus J, Townsend S, Livermore A, Wallis SC, Abdul-Aziz MH, Roberts JA. 2021. Pharmacokinetics of fluconazole and ganciclovir as combination antimicrobial chemotherapy on ECMO: a case report. *Int J Antimicrob Agents* 58:106431. <https://doi.org/10.1016/j.ijantimicag.2021.106431>
 22. Shekar K, Roberts JA, Welch S, Buscher H, Rudham S, Burrows F, Ghassabian S, Wallis SC, Levkovich B, Pellegrino V, McGuinness S, Parke R, Gilder E, Barnett AG, Walsham J, Mullany DV, Fung YL, Smith MT, Fraser JF. 2012. ASAP ECMO: antibiotic, sedative and analgesic pharmacokinetics during extracorporeal membrane oxygenation: a multi-centre study to Optimise drug therapy during ECMO. *BMC Anesthesiol* 12:29. <https://doi.org/10.1186/1471-2253-12-29>
 23. Downes KJ, Goldman JL. 2012. Too much of a good thing: defining antimicrobial therapeutic targets to minimize toxicity. *Clin Pharmacol Ther* 109:905–917. <https://doi.org/10.1002/cpt.2190>
 24. Anaissie EJ, Kontoyiannis DP, Huls C, Vartivarian SE, Karl C, Prince RA, Bosso J, Bodey GP. 1995. Safety, plasma concentrations, and efficacy of high-dose fluconazole in invasive mold infections. *J Infect Dis* 172:599–602. <https://doi.org/10.1093/infdis/172.2.599>
 25. Mulwijk EW, de Lange DW, Schouten JA, Wasmann RE, Ter Heine R, Burger DM, Colbers A, Haas PJ, Verweij PE, Brüggemann RJ. 2020. Suboptimal dosing of fluconazole in critically ill patients: time to rethink dosing. *Antimicrob Agents Chemother* 64:e00984–20. <https://doi.org/10.1128/AAC.00984-20>
 26. Sandaradura I, Marriott DJE, Day RO, Norris RLG, Pang E, Stocker SL, Reuter SE. 2021. Current fluconazole treatment regimens result in underdosing of critically ill adults during early therapy. *Eur J Clin Microbiol Infect Dis* 40:1521–1528. <https://doi.org/10.1007/s10096-021-04201-w>
 27. Watt KM, Benjamin DK, Cheifetz IM, Moorthy G, Wade KC, Smith PB, Brouwer KLR, Capparelli EV, Cohen-Wolkowicz M. 2012. Pharmacokinetics and safety of fluconazole in young infants supported with extracorporeal membrane oxygenation. *Pediatr Infect Dis J* 31:1042–1047. <https://doi.org/10.1097/INF.0b013e31825d3091>
 28. Watt KM, Gonzalez D, Benjamin DK, Brouwer KLR, Wade KC, Capparelli E, Barrett J, Cohen-Wolkowicz M. 2015. Fluconazole population pharmacokinetics and dosing for prevention and treatment of invasive candidiasis in children supported with extracorporeal membrane oxygenation. *Antimicrob Agents Chemother* 59:3935–3943. <https://doi.org/10.1128/AAC.00102-15>
 29. Krychtiuk KA, Vrints C, Wojta J, Huber K, Speidl WS. 2022. Basic mechanisms in cardiogenic shock: part 1—definition and pathophysiology. *Eur Heart J Acute Cardiovasc Care* 11:356–365. <https://doi.org/10.1093/ehjacc/zuac021>
 30. Cheng V, Abdul-Aziz M-H, Roberts JA, Shekar K. 2018. Optimising drug dosing in patients receiving extracorporeal membrane oxygenation. *J Thorac Dis* 10:S629–S641. <https://doi.org/10.21037/jtd.2017.09.154>
 31. Alobaid AS, Wallis SC, Jarrett P, Starr T, Stuart J, Lassig-Smith M, Mejia JLO, Roberts MS, Sinnollareddy MG, Roger C, Lipman J, Roberts JA. 2016. Effect of obesity on the population pharmacokinetics of fluconazole in critically ill patients. *Antimicrob Agents Chemother* 60:6550–6557. <https://doi.org/10.1128/AAC.01088-16>
 32. Martin-Loeches I, Antonelli M, Cuenca-Estrella M, Dimopoulos G, Einav S, De Waele JJ, Garnacho-Montero J, Kanj SS, Machado FR, Montravers P, Sakr Y, Sanguinetti M, Timsit J-F, Bassetti M. 2019. ESICM/ESCMID task force on practical management of invasive candidiasis in critically ill patients. *Intensive Care Med* 45:789–805. <https://doi.org/10.1007/s00134-019-05599-w>
 33. Sinnollareddy MG, Roberts MS, Lipman J, Robertson TA, Peake SL, Roberts JA. 2015. Pharmacokinetics of fluconazole in critically ill patients with acute kidney injury receiving sustained low-efficiency dialysis. *Int J Antimicrob Agents* 45:192–195. <https://doi.org/10.1016/j.ijantimicag.2014.08.013>
 34. Van Daele R, Wauters J, Lagrou K, Denooz R, Hayette M-P, Gijzen M, Brüggemann RJ, Debaveye Y, Spriet I. 2021. Pharmacokinetic variability and target attainment of fluconazole in critically ill patients. *Microorganisms* 9:2068. <https://doi.org/10.3390/microorganisms9102068>
 35. Roberts JA, Bellomo R, Cotta MO, Koch BCP, Lyster H, Ostermann M, Roger C, Shekar K, Watt K, Abdul-Aziz MH. 2022. Machines that help machines to help patients: optimising antimicrobial dosing in patients receiving extracorporeal membrane oxygenation and renal replacement therapy using dosing software. *Intensive Care Med* 48:1338–1351. <https://doi.org/10.1007/s00134-022-06847-2>
 36. Bienvenu AL, Pradat P, Matusik E, Piriou V, Rimmelé T, Parant F, Tod M, Leboucher G, Goutelle S, Antifungal study group. 2023. Suboptimal exposure to fluconazole in critically ill patients: pharmacokinetic analysis and determinants. *Infect Dis Now* 53:104630. <https://doi.org/10.1016/j.idnow.2022.10.002>
 37. Neely MN, van Guilder MG, Yamada WM, Schumitzky A, Jelliffe RW. 2012. Accurate detection of outliers and subpopulations with Pmetrics, a nonparametric and parametric pharmacometric modeling and simulation package for R. *Ther Drug Monit* 34:467–476. <https://doi.org/10.1097/FTD.0b013e31825c4ba6>
 38. EUCAST: MIC determination. 2019. Available from: http://www.eucast.org/ast_of_bacteria/mic_determination